attention_self (ViT patch self-attention): per-instruction breakdown   80 timed rows + 0 dispatcher-only CSR flips; QK body aggregates 512 repeats, PV body 128
total T1 cycles = 60060.2k   total wall = 1,006,085us   libt1 sliver = 5,081us (@60MHz)

stage assembly	group	cycles	wall
0: im2col vle8 idxH1 x2	im2col	27.8k	465us
1: im2col vle8 idxV x2	im2col	27.7k	464us
2: im2col vle8 idxH2 x2	im2col	27.6k	462us
3: im2col vle8 src x2	im2col	27.6k	462us
4: im2col vrgather H1 x2	im2col	44.4k	742us
5: im2col vrgather V x2	im2col	84.5k	1,409us
6: im2col vrgather H2 x2	im2col	63.7k	1,064us
7: im2col vse8 T x2	im2col	25.7k	431us
8: transpose vle8(V) x2	transpose	27.2k	454us
9: transpose vse8(H) x2	transpose	26.3k	441us
10: attn vle8 Q x2	QK pre	27.6k	462us
11: attn vid x2	QK pre	26.0k	435us
12: attn vmseq x2	QK pre	11.1k	187us
13: attn vmv0 x2	QK pre	27.3k	457us
14: attn vle8 Kt x4	QK pre	55.6k	930us
15: attn vmv sczero x4	QK pre	54.6k	913us
16: QK vrgather.vx K[j] x512	QK x512	7392.2k	123,617us
17: QK vwmulu.vv x512	QK x512	2739.0k	46,065us
18: QK vwredsumu.vs x512	QK x512	5933.6k	99,310us
19: QK vnsrl>>S x512	QK x512	2529.3k	42,570us
20: QK vminu255 x512	QK x512	1881.4k	31,769us
21: QK vnsrl->u8 x512	QK x512	2528.9k	42,562us
22: QK vslideup x512	QK x512	6342.8k	106,130us
23: QK vmerge x512	QK x512	7315.8k	122,347us
24: QK vmv x512	QK x512	7052.5k	117,960us
25: attn vmv COPYSA x2	softmax	27.6k	461us
26: sm vredmaxu Sa x2	softmax	18.6k	312us
27: sm vredmaxu both x2	softmax	18.7k	313us
28: sm vrgather m x2	softmax	28.9k	483us
29: sm vle8 expLUT x2	softmax	29.0k	485us
30: sm vssubu m-Sa x2	softmax	9.4k	159us
31: sm vminu127 x2	softmax	9.1k	153us
32: sm vrgather ea x2	softmax	49.4k	826us
33: sm vssubu m-Sb x2	softmax	9.4k	158us
34: sm vminu127 x2	softmax	9.1k	153us
35: sm vrgather eb x2	softmax	52.1k	870us
36: sm vwredsumu Z(a) x2	softmax	24.0k	401us
37: sm vwredsumu Z(b) x2	softmax	24.0k	402us
38: sm vwmulu Z32 x2	softmax	9.1k	153us
39: nw vnsrl idx8 x2	Newton	9.9k	166us
40: nw vle8 seedLUT x2	Newton	28.6k	478us
41: nw vrgather seed8 x2	Newton	84.4k	1,409us
42: nw vwmulu seed16 x2	Newton	9.1k	153us
43: nw vwmulu R0_32 x2	Newton	9.1k	153us
44: nw vmul i1 x2	Newton	7.8k	132us
45: nw vrsub i1 x2	Newton	7.4k	124us
46: nw vmul i1 x2	Newton	7.8k	132us
47: nw vsrl i1 x2	Newton	7.8k	132us
48: nw vmul i2 x2	Newton	7.9k	133us
49: nw vrsub i2 x2	Newton	7.3k	124us
50: nw vmul i2 x2	Newton	7.8k	132us
51: nw vsrl i2 x2	Newton	7.8k	132us
52: nw vmul i3 x2	Newton	7.8k	132us
53: nw vrsub i3 x2	Newton	7.3k	124us
54: nw vmul i3 x2	Newton	7.8k	132us
55: nw vsrl i3 x2	Newton	7.8k	132us
56: norm vnsrl R16 x2	norm	9.9k	166us
57: norm vrgather Rb16 x2	norm	33.0k	551us
58: norm vwmulu e16a x2	norm	12.5k	210us
59: norm vmul P16a x2	norm	12.9k	217us
60: norm vnsrl pqa x2	norm	32.7k	547us
61: norm vwmulu e16b x2	norm	12.5k	210us
62: norm vmul P16b x2	norm	12.9k	217us
63: norm vnsrl pqb x2	norm	32.7k	547us
64: pv vle8 Va x2	PV pre	28.3k	474us
65: pv vle8 Vb x2	PV pre	27.9k	466us
66: pv vmv Oacc0 x2	PV pre	27.3k	457us
67: PV vrgather Va col-d x128	PV x128	1848.2k	30,907us
68: PV vwmulu pqa*Va x128	PV x128	815.2k	13,689us
69: PV vwredsumu accA x128	PV x128	2269.1k	37,922us
70: PV vrgather Vb col-d x128	PV x128	1848.1k	30,905us
71: PV vwmulu pqb*Vb x128	PV x128	815.3k	13,691us
72: PV vwredsumu chain x128	PV x128	2269.1k	37,922us
73: PV vnsrl>>O x128	PV x128	632.7k	10,649us
74: PV vminu255 x128	PV x128	470.3k	7,941us
75: PV vnsrl->u8 x128	PV x128	632.1k	10,639us
76: PV vslideup x128	PV x128	1063.3k	17,825us
77: PV vmerge x128	PV x128	1172.8k	19,650us
78: PV vmv x128	PV x128	1105.9k	18,536us
79: attn vse8 O x2	store	23.6k	394us